

Sunil kumar Kommineni

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Professional Summary

Data Analyst with 2+ years of experience in data analysis, machine learning, and dashboard development using Python, SQL, and BI tools (Tableau, Power BI). Proven ability to clean, analyze, and model large datasets, build predictive models, and deliver automated reporting to support data-driven decision-making. Strong background across finance, healthcare, and education with expertise in data wrangling, generative AI, statistical modeling, and predictive analytics.

Skills

Programming & Databases: Python, SQL, R, MongoDB

Data Analysis & Statistics: Data Cleaning, EDA, KPI Reporting, Statistical Analysis, Forecasting, A/B Testing, Time Series Analysis, Reporting & Dashboards

Machine Learning: Regression, Classification, Clustering, Feature Engineering, Hyperparameter Tuning, Model Evaluation

Deep Learning & AI: TensorFlow, Keras, PyTorch, Generative AI, LLM, RAG, Prompt Engineering

Visualization & BI: Tableau, Power BI (DAX), Microsoft Excel (Pivot Tables, Power Query, VBA), Matplotlib

Data Engineering: ETL Pipelines, Data Modeling, Data Warehousing, Data Integration

Tools & Cloud: AWS, Azure, Git, Jira, Looker

Work Experience

Data Analyst | CDF Aug 2025 – Present

- Analyzed large-scale operational datasets using Python (Pandas, NumPy) and optimized SQL queries, uncovering trends that improved reporting accuracy and business decision-making.
- Designed and automated ETL pipelines, improving data processing efficiency and reducing manual reporting efforts.
- Improved data quality by identifying inconsistencies, implementing validation checks, and collaborating with stakeholders to align reporting KPIs.

Graduate Assistant – Data Analyst | University of Houston Nov 2023 – May 2025

- Analyzed and optimized scheduling datasets for 5,000+ students/semester using Excel, Python and SQL improving exam schedule accuracy and reducing conflicts by 30%.
- Designed Power BI dashboards and implemented DAX calculations to standardize exam record data, improving data quality and cross-department reporting consistency.
- Led and trained a team of 20 Graduate Assistants, establishing QA workflows and weekly reporting templates to streamline exam operations.

Projects

Bank Loan Performance Analysis and Risk Profiling

- Improved risk assessment accuracy by analyzing 10,000+ loan applications using Python and SQL, evaluating DTI, interest rate trends, and repayment behavior.
- Reduced high-risk loan approvals by 30% by implementing automated data-cleaning scripts and rule-based risk logic using Python and SQL.
- Enabled faster decision-making by developing an interactive Tableau dashboard visualizing loan status, borrower risk segments, and performance metrics.

Drug Consumption Analysis & Prediction

- Cleaned and transformed 2,000+ drug consumption records, addressing missing values, outliers and inconsistent entries to create an analysis-ready dataset.
- Conducted exploratory data analysis using 15+ visualizations including distribution plots, correlation heatmaps and factor analysis to identify behavioral patterns.
- Constructed predictive models using Logistic Regression, Random Forest, and SVM achieving 75% accuracy and elevated model performance through hyperparameter tuning considering various model metrics.

LLM-Powered Document Question Answering System

- Built a RAG-based QA system using LLM embeddings and semantic search, enabling accurate answers from 100+ documents.
- Implemented text chunking, retrieval, reducing response time by 40% and improving answer relevance.
- Enhanced model reliability by supplying retrieved context to the LLM, reducing hallucinations by 25% and maintaining 90%+ answer accuracy on test queries.

Education

University of Houston -- Houston, TX

Master's in Engineering Data Science | GPA: 4.0/4.0

May 2025

Certifications

- Oracle Cloud Infrastructure Generative AI Professional (2025)
- Microsoft Certified: Power BI Data Analyst Associate (PL-300)
- Microsoft Certified: Fabric Analytics Engineer Associate (DP-700)